LEE M. BERNARD

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Software Engineer

Software Engineer and M.S. Computer Science student who builds and ships full-stack and AI applications: a live multilingual RAG voice assistant (Python, FastAPI, Next.js, LLMs), a computer-vision chess-playing robot arm, and production websites. Co-author of an IEEE ICIICE 2026 paper on augmented reality.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, C#, HTML, CSS
Frameworks & Libraries: Next.js, React, FastAPI, Tailwind CSS, LlamaIndex, OpenCV, Unity, AR Foundation, Flutter, SFML
AI & Data: RAG, Large Language Models (LLMs), Pinecone, PostgreSQL, Supabase, Speech-to-Text, Computer Vision
Tools & Practices: Git, GitHub Actions (CI/CD), Docker, Linux, REST APIs, Unit Testing (pytest, TestNG), Selenium, Agile, OOP
Hardware & Security: Raspberry Pi, ESP32, NIST Cybersecurity Framework, GLBA, PCI-DSS | Speaks English, French, Haitian Creole

PROFESSIONAL EXPERIENCE

Lab Assistant, VIPER Lab — January 2026 – Present

Florida Gulf Coast University, U.A. Whitaker College of Engineering, Fort Myers, FL

- Built a chess-playing Dobot robot arm in Python: OpenCV board vision, Stockfish on a Raspberry Pi, ESP32 touchscreen.
- Maintain and troubleshoot the lab's VR/AR systems (Oculus, NVIDIA GPUs, Windows, Steam) and set up new hardware.
- Support students and faculty with VR/AR projects and demo the lab at classes and outreach events.

Web Developer Intern — June 2024 – August 2024

Chery Health Care, North Port, FL

- Built a responsive website with a small development team using Next.js, React, TypeScript, HTML and CSS.
- Developed a custom emailing service that automates communication from clients to the company.
- Set up domain registration and a custom domain email system, and ran functionality and performance tests before launch.

Learning Hub Student Advisor — January 2023 – May 2025

Florida Gulf Coast University, U.A. Whitaker College of Engineering, Fort Myers, FL

- Tutored engineering students one-on-one and coached study habits, time management and problem-solving.
- Ran peer-learning workshops and built resources for new mentors; referred students to campus support.

PROJECTS

Ask the Eagle: Multilingual AI Voice Assistant — Summer 2026

Python, FastAPI, Next.js, TypeScript, Pinecone, Groq LLM, Docker | fgcu-speach-ai.vercel.app

- Built a full-stack, multilingual RAG assistant that answers students' course, advising and deadline questions by voice or text.
- Designed FastAPI REST endpoints for retrieval (Pinecone, Cohere), LLM answers (Groq), speech-to-text and text-to-speech.
- Automated a daily scraper and GitHub Actions CI/CD that re-index changed documents and run pytest unit tests.

Unite U App — August 2024 – May 2025

FlutterFlow | Team of six

- Built a campus events app for FGCU students, owning backend logic for user data, event listings and authentication.

Reddit End-to-End Test Automation — Spring 2024

Java, Selenium WebDriver, TestNG, Maven

- Wrote automated UI tests for Reddit login, post creation, search, and home, subreddit and user pages, run as TestNG suites.

More: Phishing awareness quiz (HTML, CSS, JavaScript) • Tic-Tac-Toe with AI opponent (C++/SFML, Flutter) • Autonomous rover control (C++) • More on github.com/Lerbernard

PUBLICATIONS

A. Balan, **L. Bernard** and M. B. Islam, "Augmented Reality for Corporate Sustainability Reporting: An Interactive Carbon Footprint System," 2026 IEEE International Conference on Integrated Intelligence and Cognitive Engineering (ICIICE), Dubai, UAE, pp. 1–6. doi:10.1109/ICIICE69672.2026.11565311

EDUCATION

M.S. Computer Science, Florida Gulf Coast University, Fort Myers, FL — Expected December 2026
B.S. Software Engineering, Florida Gulf Coast University, Fort Myers, FL — 2025